

# **The Invisible Elephant in the Galaxy**

## **Why dark-matter halos are a patch, not an answer — and why galaxies keep pointing at dark energy instead**

Carl P. Zimmerman · Briar Creek Tech

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Picture a merry-go-round at the park. If it spins too fast, the kids on the edge get flung off. The faster it spins, the harder you have to hold on. Now look up at a spiral galaxy. The stars on the edge are racing around the center so fast that, by every rule we know, they should be flung out into space. But they aren't. They keep circling, calm as can be, year after year for billions of years.

Something is holding them in. The question that has split physics for fifty years is simple to ask and brutally hard to answer: **what?**

There are only two honest options. Either there is a huge amount of invisible stuff we cannot see, adding extra gravity — that is the “dark matter” answer. Or our rule for how gravity works is slightly wrong when gravity gets very weak — that is the “modified gravity” answer. This editorial is about why the first answer, after fifty years and billions of dollars, looks less and less like science and more and more like a security blanket — and why a newer idea, which I will call the de Sitter answer, deserves a serious look.

### **The thing nobody can find**

Let me be fair to dark matter first, because it started as a perfectly reasonable idea. Galaxies spin too fast, so maybe there is extra matter we can't see — cold, dark, silent stuff spread out in a giant invisible ball, or “halo,” around each galaxy. Add enough of it in the right places and the math works. Fine.

Here is the problem. We have been hunting for that invisible stuff for half a century. We built detectors a mile underground, shielded from everything, waiting for one tiny ping of a dark-matter particle hitting a normal atom. We smashed particles together in the biggest machines ever built, hoping to make some. We have looked, and looked, and looked.

We have found **nothing**. Not one particle. Every year the detectors get more sensitive, the net gets finer, and every year the net comes up empty. At some point you have to ask the uncomfortable question: if a thing is invisible, undetectable, and never shows up no matter how hard you look — how is that different from a thing that isn't there?

Dark matter has become the friend who always cancels. "I swear I was going to come, something came up." Maybe. But after fifty cancellations, you stop setting a place at the table.

## **The conspiracy nobody likes to talk about**

The missing particle is bad. The next problem is worse, and it is the one that turned me from a skeptic into someone who thinks the whole halo idea is upside down.

When astronomers measured thousands of galaxies — big ones, small ones, fat spirals, faint dwarfs — they found something that should not exist if dark matter is real. They found that **wherever the normal, visible matter is, the total gravity is completely predictable**. You tell me how much ordinary stuff a galaxy has and where it sits, and I can tell you exactly how fast its stars will spin — including all the "extra" pull that's supposed to come from the invisible halo. Every galaxy. The relationship is so tight it's almost a law of nature. Scientists call it the radial acceleration relation. I call it the smoking gun.

Think about what that means. The visible matter and the supposedly invisible matter are perfectly in step, everywhere, in every galaxy, as if they were following the same conductor. But dark matter isn't supposed to care about ordinary matter. They barely talk to each other — that's the whole point of dark matter being "dark." So how do two things that ignore each other end up marching in perfect lockstep across the entire universe?

The honest answer is: they shouldn't. To make halos reproduce that tight relationship, you have to tune each invisible halo by hand to match the visible galaxy it surrounds — a different custom-fit halo for every single galaxy, each one arranged just so. Physicists have a word for needing to secretly arrange things to get the answer you already know: a **fine-tuning conspiracy**. It's the kind of thing that, in any other field, makes you suspect you've got the wrong idea.

And here's the tell. That tight relationship doesn't kick in at just any strength of gravity. It kicks in at one specific value — a particular gentleness of pull, the same number in every galaxy. Stars feeling gravity stronger than that number behave normally. Stars feeling gravity weaker than that number start needing the "extra." There is a line in the sand, a single special acceleration, written into every galaxy in the sky.

Dark matter has no reason to know about that number. It's just supposed to be invisible stuff sitting wherever gravity dumped it. Why would invisible stuff, scattered randomly, draw the exact same line in every galaxy? It's like discovering that every vending machine

on Earth, built by different companies on different continents, charges exactly the same secret price for a hidden item. That's not a coincidence. That's a clue that something deeper is going on — something the machines all share.

## Following the clue

So what could every galaxy in the universe possibly share? Not invisible matter — that's different in every galaxy. The thing they all share is the universe itself. The empty space they all float in. And empty space, it turns out, is not empty.

Here is the part that gives me chills. We've known since the 1990s that the universe is not just expanding — it's speeding up, pushed apart by something woven into empty space. We call it dark energy, and we measure its strength with a number called the cosmological constant. It is, as far as we can tell, the same everywhere. It's the one thing every galaxy truly has in common.

Now take that special acceleration written into every galaxy — the line in the sand where stars start misbehaving. Work out, from the strength of dark energy alone, what acceleration empty space should set. You get a number. Compare it to the line in the sand. **They're the same number.** Roughly a ten-billionth of the gravity you feel standing on Earth, give or take, the galaxy line and the dark-energy number land right on top of each other.

Let that sink in. The mysterious acceleration that decides where galaxies need “extra” gravity is the same as the acceleration set by the dark energy pushing the whole universe apart. Dark matter calls that a coincidence — a meaningless accident, two unrelated numbers that just happen to match. After everything else, I don't believe in that coincidence anymore. I think it's the universe telling us, in plain language, where the answer is.

## The de Sitter answer

That's the idea I've spent years chasing. In one line:

$$a_0 = c^2 \sqrt{\frac{\Lambda}{32\pi}} \approx 9.36 \times 10^{-11} \text{ meters per second, per second.}$$

Don't worry about the symbols. Here's the sentence: the galaxy line ( $a_0$ ) is set by the dark energy of empty space ( $\Lambda$ ). One short formula. No invisible particles. No custom halos. The same dark energy that accelerates the universe is the thing that quietly changes how gravity and motion work when the pull gets very gentle.

Why would empty space do that? Because empty space, in a universe with dark energy, has a faint built-in warmth — a tiny temperature you can't turn off, a leftover hum from the expansion itself. When something accelerates hard, it doesn't notice that hum. When something drifts almost in free fall — like a star in the lonely outskirts of a galaxy —

the hum becomes a real part of how much it resists being pushed. That gentle resistance is exactly what looks, from the outside, like “extra gravity.” It isn’t extra matter. It’s the universe’s own background coming through.

I want to be honest about what is and isn’t nailed down, because that honesty is the whole point. The shape of this law — that the galaxy line comes from dark energy — is forced by the math in several independent ways, and a computer-checked proof backs up the spine of it. What’s *not* fully pinned down is one plain number out front, a simple factor of one-half, which the theory takes as its single honest assumption. That’s it. One number. Compare that to dark matter, which needs a brand-new invisible particle that no experiment has ever seen, plus a custom-built halo for every galaxy in the sky. One honest assumption versus an entire invisible universe arranged by hand. I know which one I find simpler.

## **The part dark matter can’t do**

Now here’s the argument that, for me, ends the debate — and it’s not about who sounds smarter. It’s about who is willing to be wrong.

A good scientific idea sticks its neck out. It says, “if you go measure *this*, you will see *that* — and if you don’t, I’m finished.” Dark matter, sadly, has stopped doing this. It has so many adjustable knobs — how much invisible stuff, where it sits, how it clumps — that almost any result can be absorbed. Galaxies too fast? Add halo. Too slow? Less halo. It bends to fit. A theory that can explain any outcome has stopped predicting anything.

The de Sitter idea cannot hide like that, because it has almost nothing to hide behind. It makes two flat predictions you can go check:

**One.** If the galaxy line comes from dark energy, and dark energy was different long ago, then the line should have been different long ago too. Specifically, look at galaxies as they were when the universe was a quarter of its current age. The theory says their special acceleration should be lower — about three-quarters of today’s value. Dark matter says it should be the same, or even higher. Those are wildly different answers, and the giant telescopes coming online this decade can tell them apart. If the line doesn’t shrink the way the theory says, the theory is wrong. Done.

**Two.** Because this is a change in how *motion* works — not a cloud of invisible stuff — it switches off completely where gravity is strong, like inside our own solar system. We already tested that. The Cassini spacecraft, on its way to Saturn, let us measure the Sun’s gravity with breathtaking precision. The de Sitter idea predicts the effect there is essentially zero, hidden far below what Cassini could see — and that’s exactly what Cassini found. A rival class of “modified gravity” theories predicted a wobble Cassini should have

caught. It wasn't there. The de Sitter idea passed a test that has already killed some of its competitors.

That's the difference. One idea makes risky bets and so far keeps winning them. The other rearranges invisible furniture to match whatever we find. Only one of those is doing science.

## What I'm actually asking for

I'm not asking you to believe galaxies have no dark matter for certain. I can't prove a negative, and I won't pretend to. The universe might still surprise us; a particle might still show up tomorrow. If  $a_0$  doesn't shrink in the early universe the way the theory says, I'll tell you the idea failed, and I'll mean it.

What I'm asking is smaller and, I think, harder to argue with: **stop treating the invisible answer as the safe, default, grown-up choice**. It isn't. It's an enormous, undetectable, hand-tuned assumption that has come up empty for fifty years and quietly ignores the loudest clue in the data — the fact that galaxies and dark energy share the exact same magic number.

The simpler idea is sitting right in front of us. It needs no new particles. It explains the one number dark matter calls a coincidence. It ties the spinning of galaxies to the stretching of the whole universe with a single short formula. And — unlike its rival — it is brave enough to tell you exactly how to prove it wrong.

In science, the theory that sticks its neck out deserves the benefit of the doubt over the one that hides. Dark-matter halos have been hiding for fifty years. It's time to follow the clue the galaxies have been handing us the whole time. It points straight at the dark energy in the empty space between the stars — and that is where I'm betting the answer is.

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*This is an opinion piece. The case against dark-matter halos (the missing particle, the fine-tuning of the radial acceleration relation, the shared acceleration scale) reflects genuine, published criticisms of the paradigm. The de Sitter framework's strengths (tying  $a_0$  to the cosmological constant, the radial-acceleration and baryonic Tully–Fisher fits, the Cassini pass, the falsifiable  $a_0(z)$  prediction) are real; its open questions (the single free factor  $\kappa = \frac{1}{2}$ , the unbuilt covariant completion, the contested high-redshift direction) are equally real and stated in the technical papers. The framework is a falsifiable theory of gravity and the dark sector — not yet a proven replacement for dark matter, and not a theory of everything. DOI: 10.5281/zenodo.20721540.*